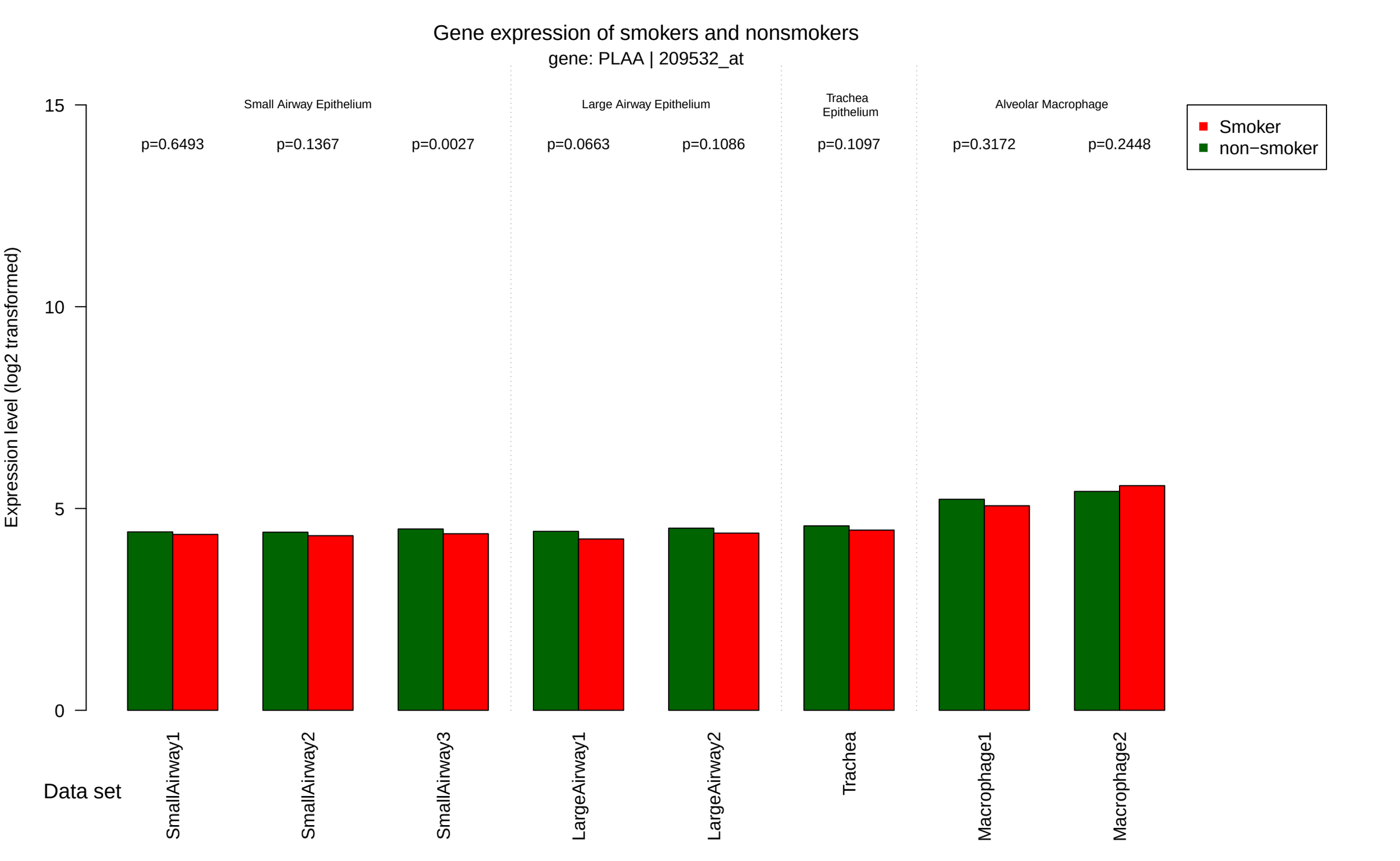
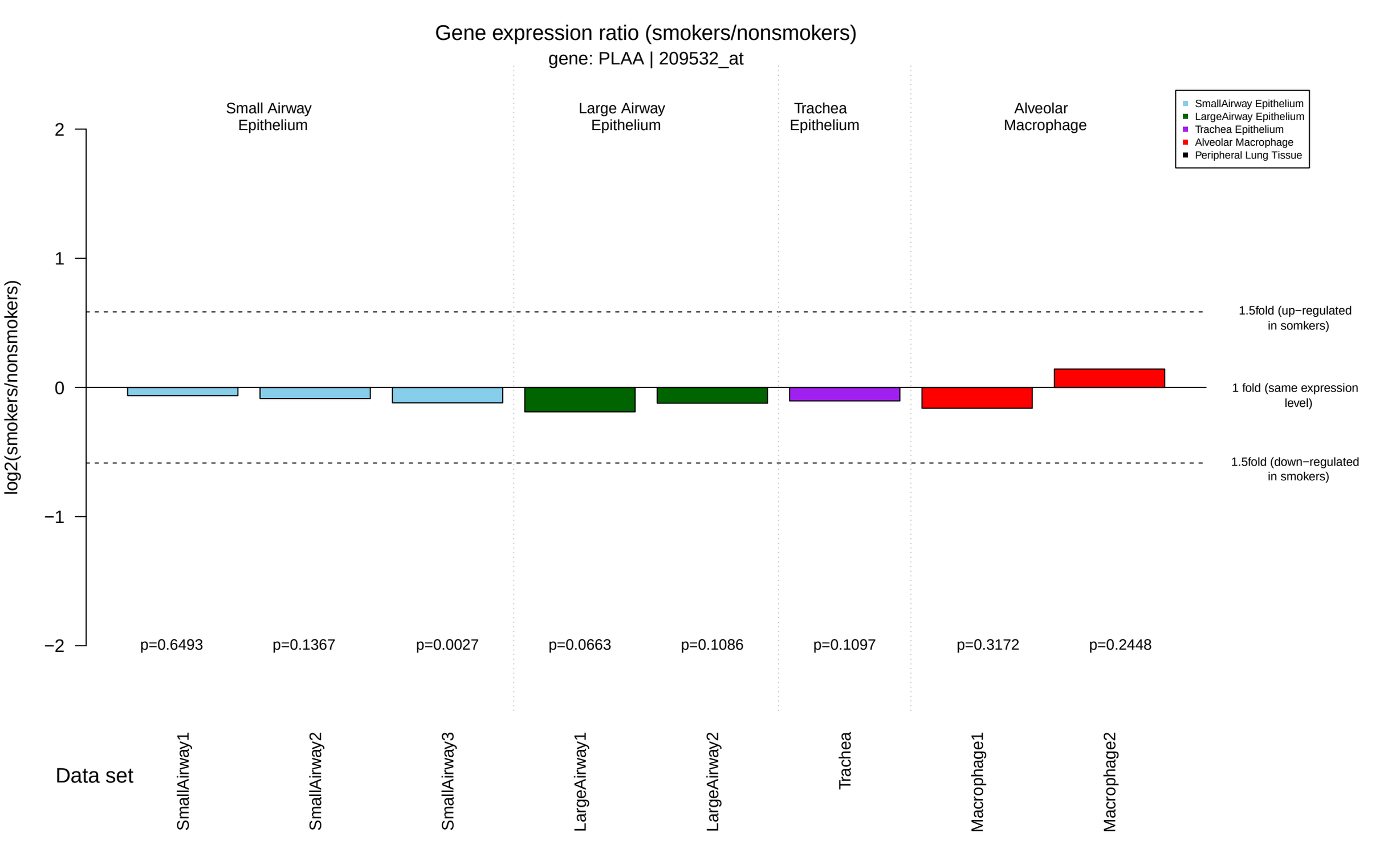
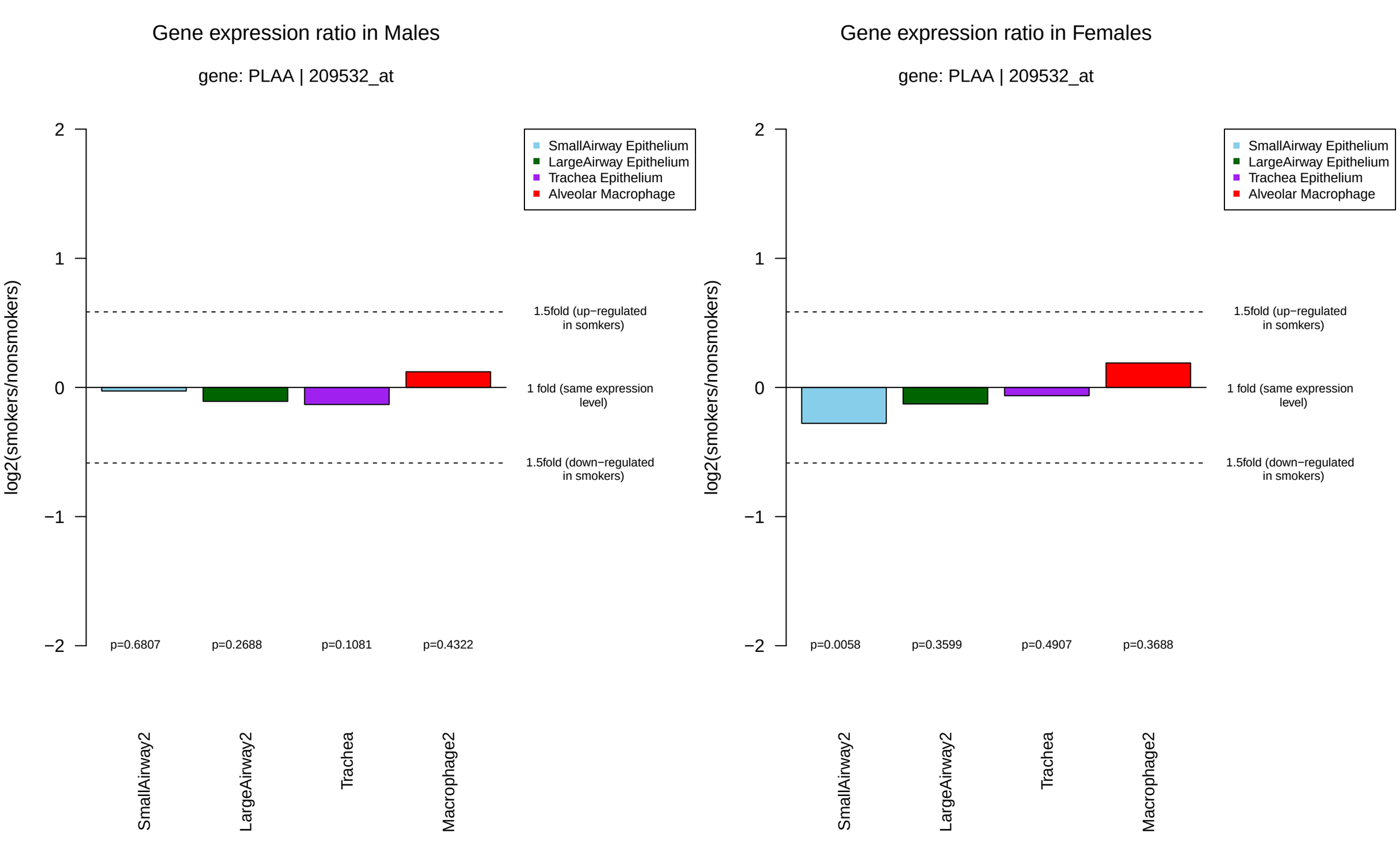
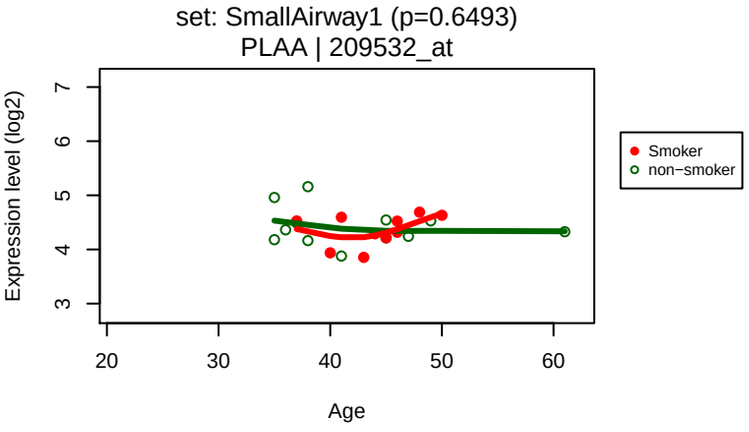
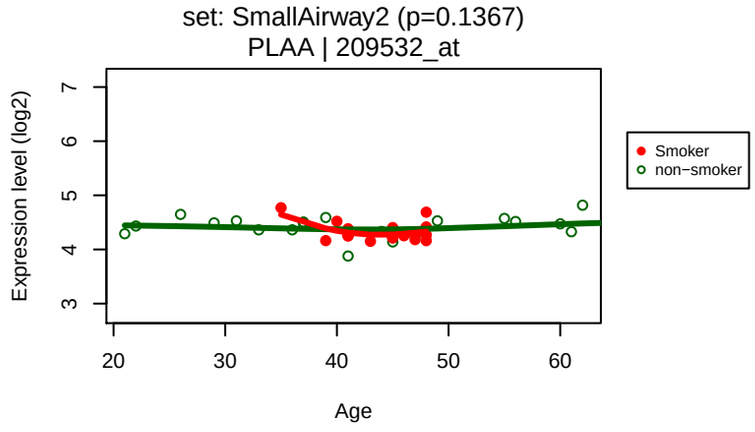
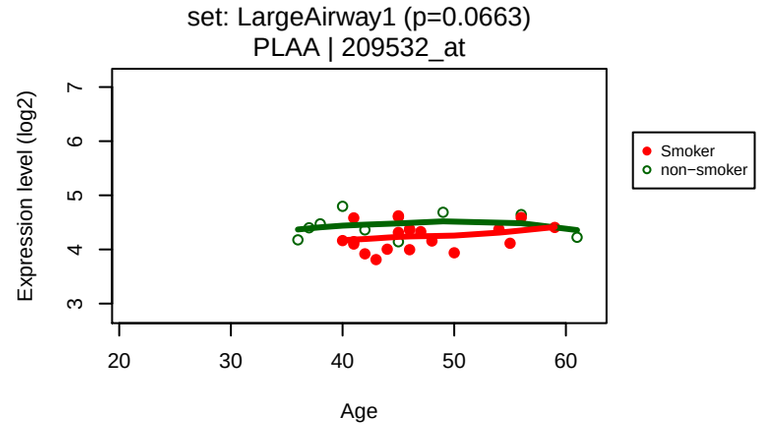

Gene expression of smokers and nonsmokers
gene: PLAA | 209532_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6493
p=0.1367
p=0.0027
p=0.0663
p=0.1086
p=0.1097
p=0.3172
p=0.2448
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

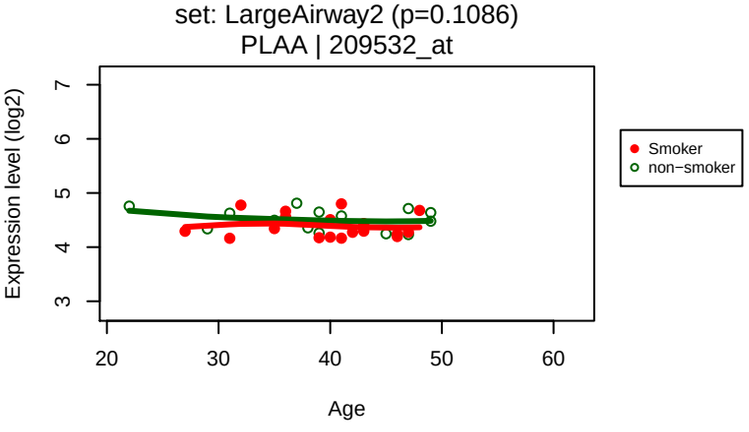

Gene expression ratio (smokers/nonsmokers)
gene: PLAA | 209532_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6493
p=0.1367
p=0.0027
p=0.0663
p=0.1086
p=0.1097
p=0.3172
p=0.2448
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

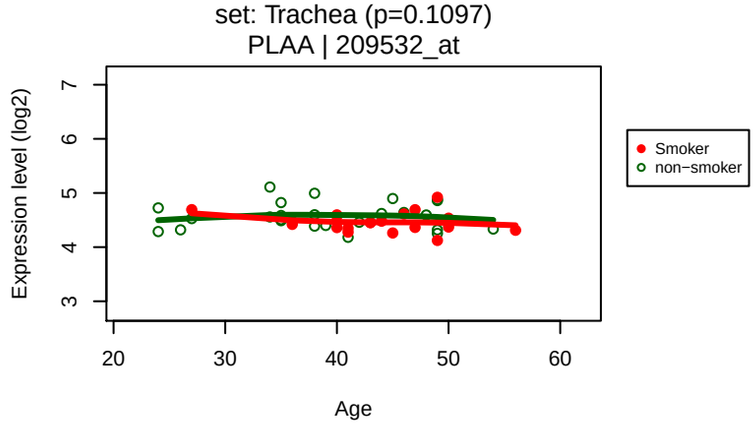

Gene expression ratio in Males
gene: PLAA | 209532_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6807
p=0.2688
p=0.1081
p=0.4322
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PLAA | 209532_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0058
p=0.3599
p=0.4907
p=0.3688
SmallAirway2
LargeAirway2
Trachea
Macrophage2

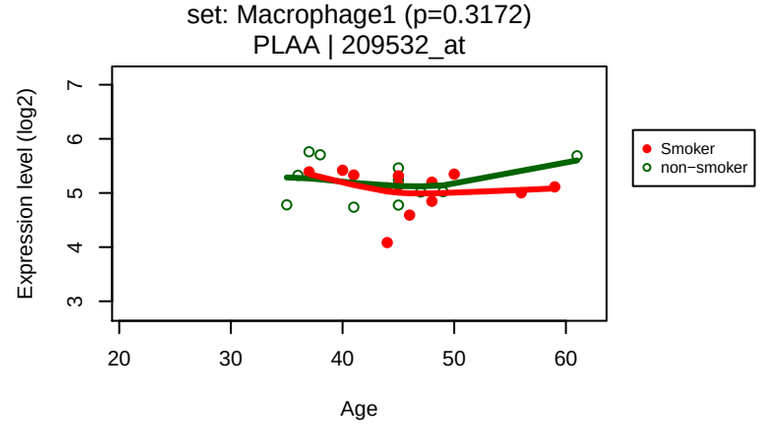

set: SmallAirway1 (p=0.6493)
PLAA | 209532_at
Expression level (log2)
Age
Smoker
non−smoker

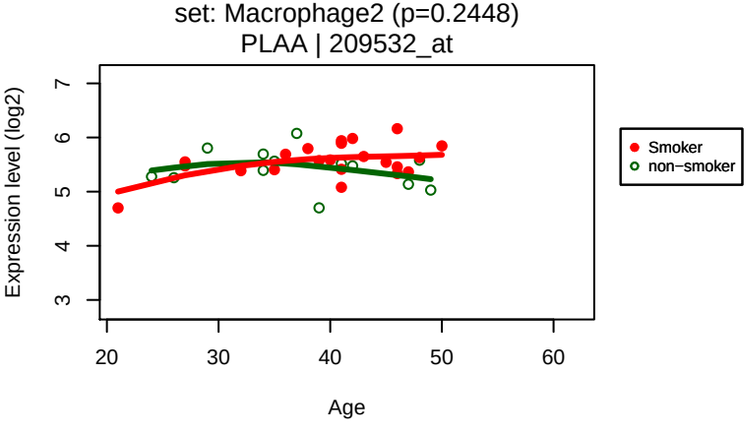

set: SmallAirway2 (p=0.1367)
PLAA | 209532_at
Expression level (log2)
Age
Smoker
non−smoker

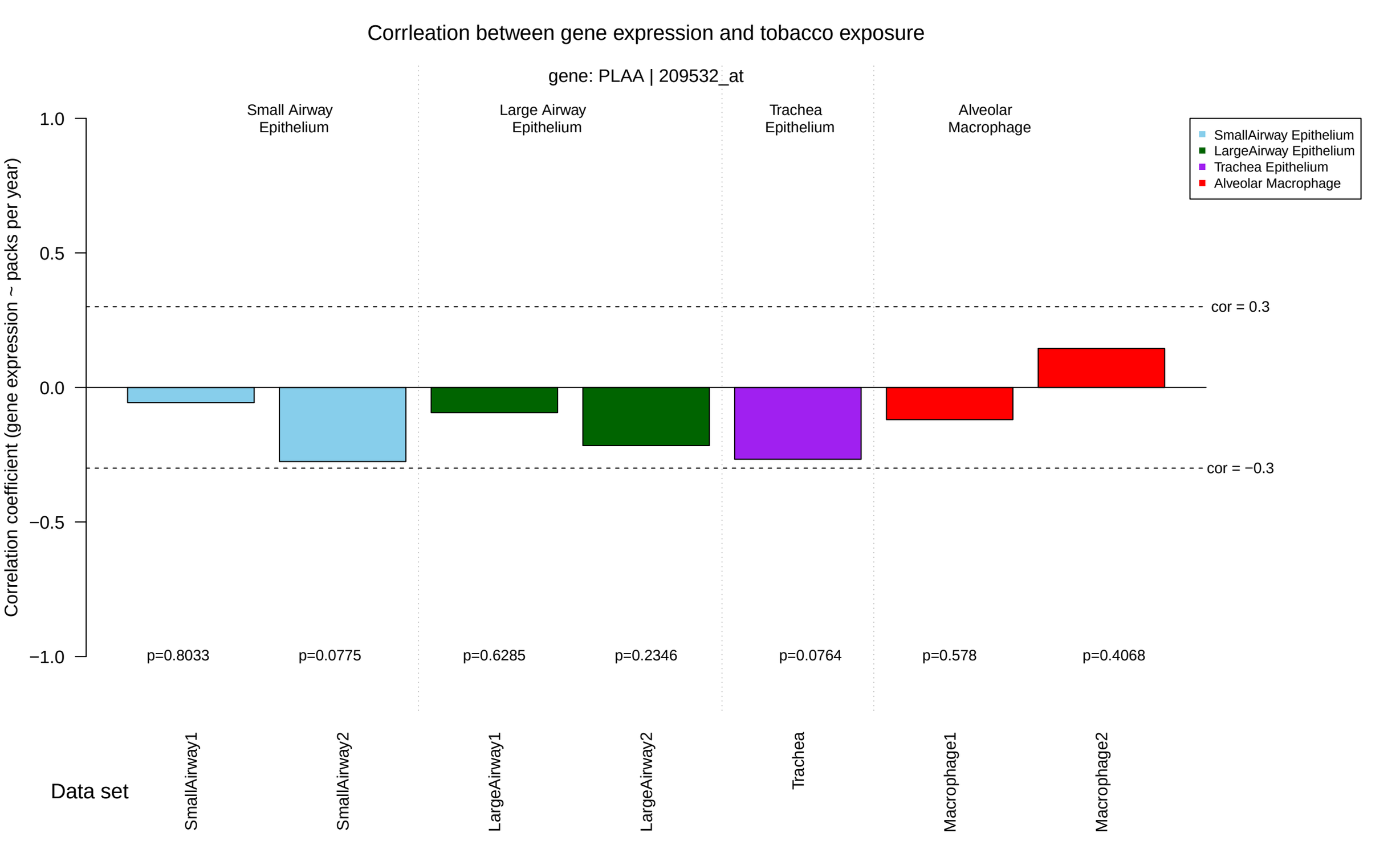

set: LargeAirway1 (p=0.0663)
PLAA | 209532_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1086)
PLAA | 209532_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1097)
PLAA | 209532_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3172)
PLAA | 209532_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2448)
PLAA | 209532_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PLAA | 209532_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.8033
p=0.0775
p=0.6285
p=0.2346
p=0.0764
p=0.578
p=0.4068
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2